

Energy Efficient Implementation of SNNs using RRAM based In-Memory Computing (IMC).

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Abstract

In recent years, Spiking Neural Networks (SNN) has emerged as a prominent candidate for resource constrained neural computing due to its inherent features of network sparsity and event driven execution. Due to these features SNNs are energy efficient, scalable, have a low memory footprint. However, conventional SNN implementations on von Neumann architectures suffer from significant energy and latency bottlenecks. In-memory computing (IMC) using Resistive Random Access Memory (RRAM) offers a potential solution by integrating computation and storage, reducing data movement overhead. This work presents an RRAM-based IMC approach for SNN implementation, leveraging a crossbar array for vector-matrix multiplication (VMM). A 3-layer SNN is trained using SNN-Torch on the MNIST dataset, and the trained weights are mapped onto the RRAM crossbar. Preliminary results include behavioral modeling of a 1T-1R RRAM cell and its I-V characteristics. Future work involves scaling to crossbar arrays, designing encoding circuits, and integrating online learning via Spike Timing Dependent Plasticity (STDP).

Origin of the problem

The increase in availability of data and advancements in storage technologies has led to an exponential increase in AI applications in various fields. These rapid advancements have intensified the need for energy-efficient computing systems capable of handling complex cognitive tasks within power and resource constraints. SNNs are biologically inspired models that have emerged as a promising alternative to traditional deep neural networks (DNNs) due to their event-driven nature and potential for high energy efficiency. However, conventional implementations of SNNs on von Neumann architectures suffer from significant energy and latency bottlenecks, primarily due to the frequent data transfer between memory and processing units (memory wall), thus negating the inherent advantages of SNNs. This bottleneck makes traditional CMOS-based implementations impractical for real-world applications requiring low-power, real-time inference and learning.

IMC architectures, particularly those based on RRAM, offer a promising solution to these issues. By integrating storage and computation within the same hardware unit, IMC reduces data movement overhead and enables efficient execution of VMM - an essential operation in DNNs. While several studies have explored IMC-based SNNs, the existing implementations often compromise on accuracy, scalability, or biological plausibility. Addressing these gaps is crucial for developing energy-efficient and scalable SNN architectures for practical neuromorphic computing applications.

Relevance in neuromorphic device technology

The development of energy-efficient neuromorphic systems is highly relevant in the context of modern memory and computing architectures, particularly as the demand for AI-driven applications grows. Neuromorphic computing, inspired by the human brain's ability to process information in parallel and with low power, stands to benefit greatly from memory-centric architectures like IMC. By eliminating the need for frequent data movement, IMC architectures significantly reduce energy consumption and latency, making them ideal for implementing energy-efficient SNNs. RRAM-based IMC, in particular, offers a promising solution due to its compatibility with CMOS technology, scalability, and ability to emulate synaptic behavior, which is critical for neuromorphic computing.

Methodology

The methodology for this project is primarily based on the development of an energy-efficient SNN using RRAM-based CIM architecture (Ref. [1], [2]). Key steps include:

1. Training and implementation of a 3-layer SNN using SNN_Torch library trained with Backpropagation Through Time (BPTT) algorithm. The MNIST dataset is used to train the network and evaluate its accuracy
2. Using available RRAM models to build a crossbar array consisting of 1T1R units modeled in Verilog-A
3. Using the trained weights obtained from trained SNN model to define conductance's of RRAMs in each cell of the crossbar and designing of an encoding circuit at the I/O interfaces of the RRAM crossbar to process the analog voltages.
4. Leverage the RRAM-based crossbar as an IMC unit to perform VMM operation for SNN execution.
5. Simulation and validation on the MNIST dataset as well as analyzing the energy consumption per spike and comparing with conventional CMOS-based implementations.

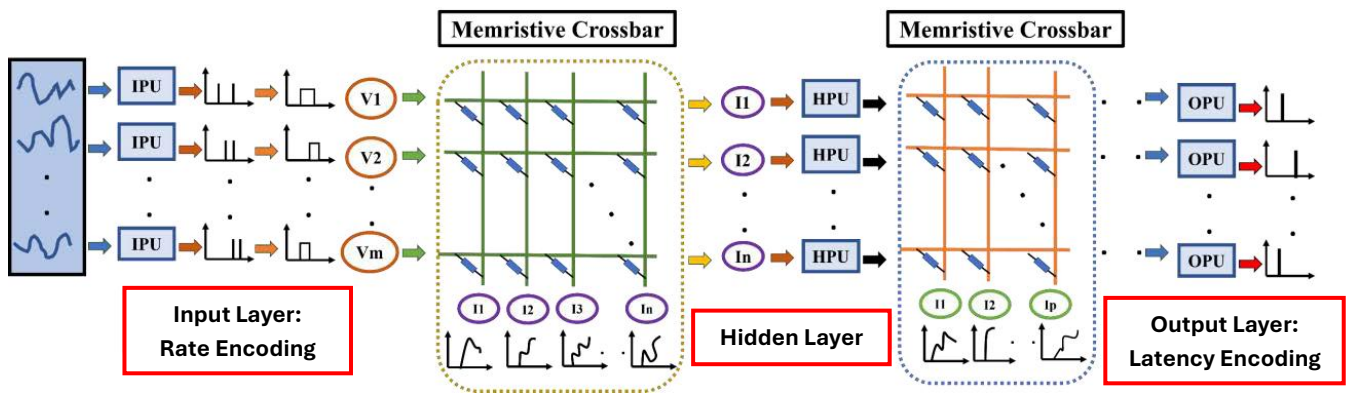


Fig: Architecture of the designed IMC-based SNN (Ref. [2])

Preliminary Results

I. RRAM

Resistive Random Access Memory (RRAM) is a non-volatile memory technology that leverages resistive switching. Here we present the behavioral modeling of an RRAM device using Verilog-A and its integration into a crossbar architecture.

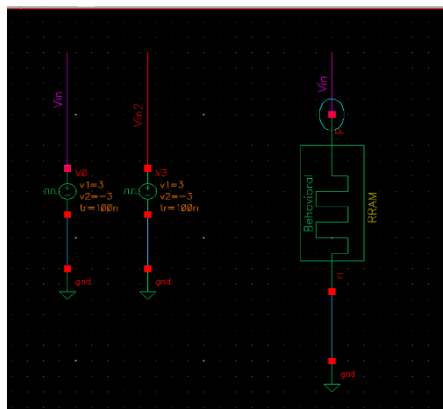
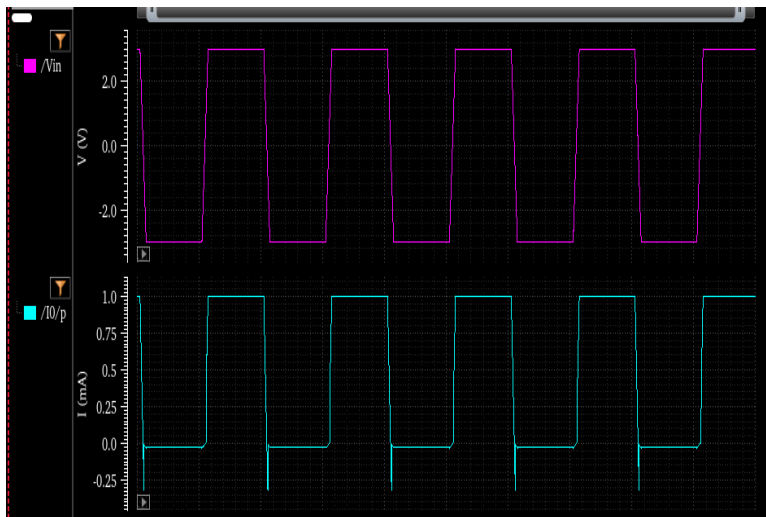
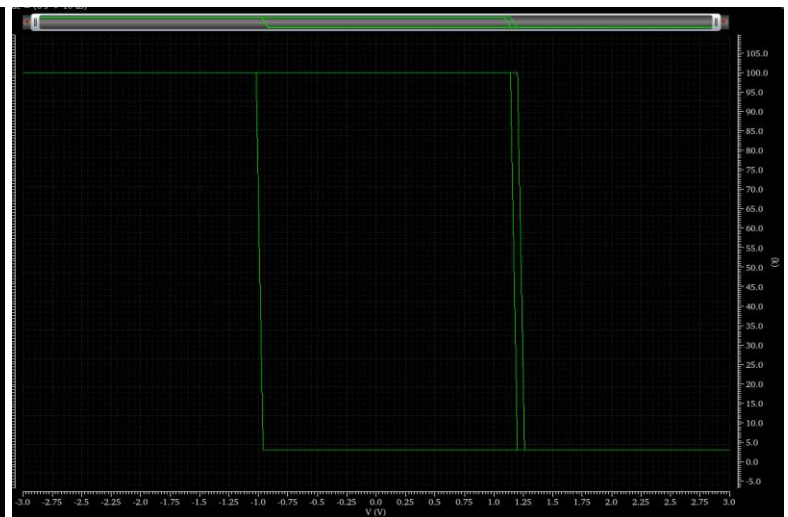


Fig: Two terminal RRAM schematic



(a)



(b)

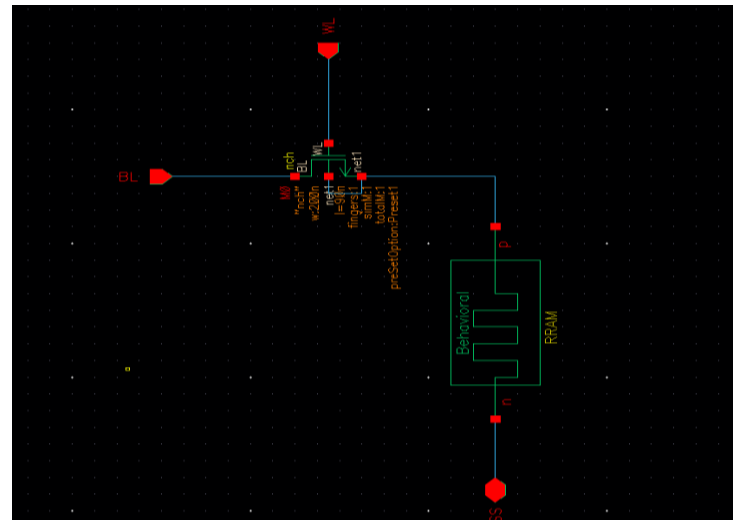
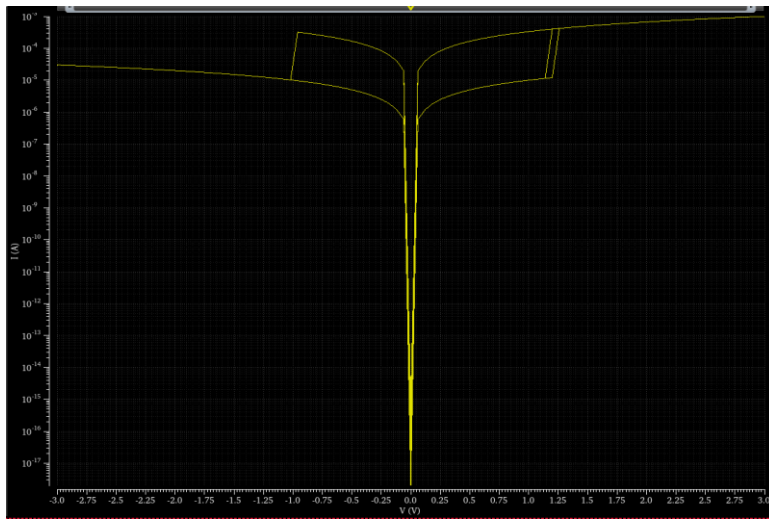


Fig: (a)(b)(c) Output characteristics: (a) Transient response, (b) Resistance v/s Voltage and (c) Current v/s Voltage. (d) 90nm Access transistor circuit

The I-V characteristics of RRAM exhibit two major resistive states:

High Resistance State (HRS): When the RRAM cell is initially in the off state, the conductive filament is not formed, and the device exhibits high resistance.

Low Resistance State (LRS): Upon applying a voltage greater than the SET threshold (V_{SET}), the conductive filament is formed, reducing the resistance.

Reset Process: When a voltage below the RESET threshold (V_{RESET}) is applied, the conductive filament dissolves, returning the device to HRS.

90nm access transistor circuit: The access transistor is designed using a 90nm technology node, $V_{dd} = 1.2$ V while the gate voltage is 1.8 V ($V_{dd} + V_{th}$) to ensure proper switching. The transistor enables controlled switching between HRS and LRS.

II. SNN Training

The SNN is implemented using the SNN-Torch library in Python and trained on the MNIST dataset for five epochs. The neuron model employed is the Leaky Integrate-and-Fire (LIF) neuron, with a membrane potential decay factor (β) set to 0.95. The spike train duration is configured to 200ms.

The SNN follows a three-layer Multi-Layer Perceptron (MLP) architecture with a structure of 196-400-10. The images in the MNIST dataset are resized to 14x14, thus 196 neurons in the first layer. The final test accuracy of the model obtained after five epochs is 96.16%.

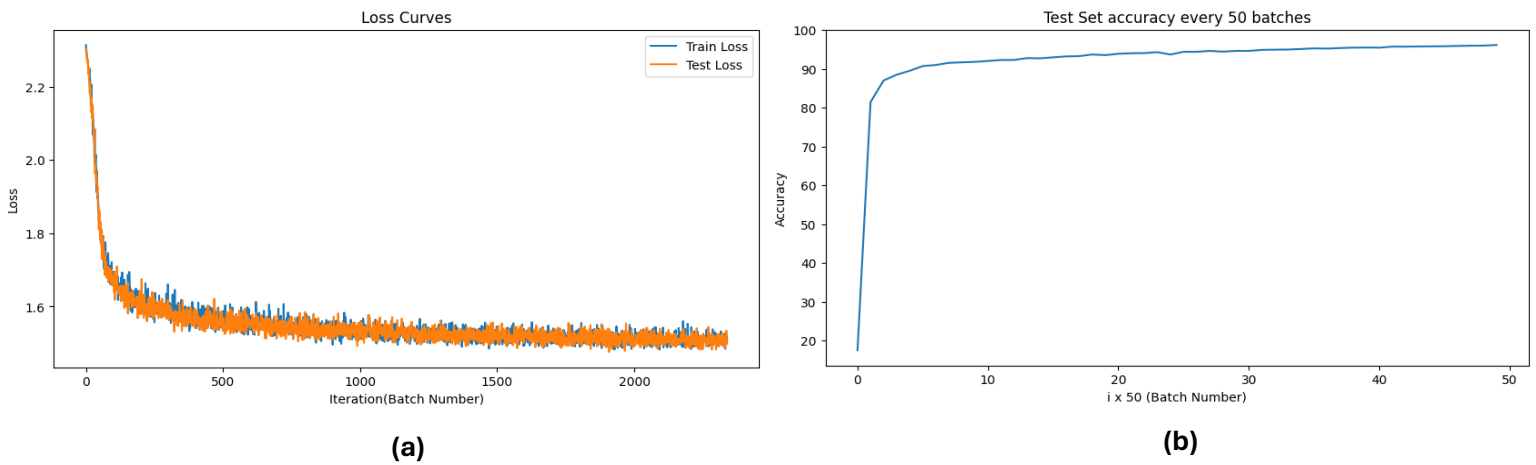


Fig: (a) Training & Testing Loss and (b) the test set accuracy of the SNN as a function of batch number

Conclusion/Future Work:

Currently, we have designed and simulated a single 1T-1R RRAM cell, along with its output characteristics. Next, our focus is on the development of RRAM crossbar arrays for IMC in SNNs. Additionally, an encoding circuit needs to be designed at the I/O interfaces of the crossbar to process analog voltages efficiently. Further, we will try to integrate online learning capabilities directly into the IMC framework, enabling real-time weight updates through the STDP algorithm. The proposed architecture will be simulated and analyzed for energy consumption per spike, with a comparative evaluation against existing implementations.

References:

- [1] El Arrassi, Asmae, et al. "Energy-efficient SNN implementation using RRAM-based computation in-memory (CIM)." *2022 IFIP/IEEE 30th International Conference on Very Large-Scale Integration (VLSI-SoC)*. IEEE, 2022.
- [2] Nowshin, Fabiha, and Yang Yi. "Memristor-based deep spiking neural network with a computing-in-memory architecture." *2022 23rd International Symposium on Quality Electronic Design (ISQED)*. IEEE, 2022.